

Abstract

We describe an abstract framework for mathematical conjecture settling by a collection of cooperating agents operating over a shared bank of verified mathematical statements.

Agents may pursue different settlement objectives, including proving a conjecture, proving its negation, or establishing its independence from a specified axiom system.

Unlike independent dovetailed proof searches, the agents communicate through a monotonically growing shared lemma bank. Every verified result produced by one agent therefore becomes available as a premise to every subsequent agent.

The framework separates the mathematical structure of conjecture settling from any particular implementation by automated theorem provers, proof assistants, search algorithms, or learned systems.

1 Formal Setting

Let A be a formal axiom system and let C be a conjecture expressed in the language of A .

We distinguish three possible settlement objectives:

$$A \vdash C,$$

$$A \vdash \neg C,$$

and

$$A \not\vdash C \quad \text{and} \quad A \not\vdash \neg C.$$

The third condition corresponds to independence of C from A , assuming the relevant consistency and metatheoretic conditions are satisfied.

Failure to discover a proof of either C or $\neg C$ is not by itself a certificate of independence.

2 The Shared Lemma Bank

Let L_t denote the collection of verified statements available after round t .

Initially,

$$L_0 = A.$$

More generally, one may distinguish background axioms B from problem-specific assumptions A , in which case

$$L_0 = B \cup A.$$

During round t , agents may derive new verified statements. Let D_t denote the set of statements successfully deposited during that round.

The bank evolves according to

$$L_{t+1} = L_t \cup D_t .$$

Consequently,

$$L_t \subseteq L_{t+1} .$$

Thus mathematical knowledge accumulated by the system is monotonic: once a verified lemma has entered the bank, it remains available to all future agents.

3 Agents

For the initial system, consider three agents:

$$G = \{P, R, I\}.$$

Their objectives are different even though they operate over the same bank.

3.1 The Prover

The prover P seeks a certificate establishing

$$A \vdash C.$$

At each activation, P may select any finite collection of statements from the current bank,

$$\Gamma_P \subseteq_{\text{fin}} L_t ,$$

apply an admissible inference procedure, and attempt to produce a new verified statement.

3.2 The Refuter

The refuter R seeks a certificate establishing

$$A \vdash \neg C.$$

It has access to the same lemma bank as the prover.

Thus a lemma discovered while attempting to prove C may later assist the search for $\neg C$, and conversely.

3.3 The Independence Agent

The independence agent I seeks a certificate establishing that C is independent of A .

Conceptually, its objective is

$$A \not\vdash C \quad \text{and} \quad A \not\vdash \neg C.$$

Unlike ordinary proof search, this generally requires metamathematical certificates.

For example, an independence argument may establish suitable models

$$M \models A \cup \{C\}$$

and

$$N \models A \cup \{\neg C\}.$$

Other implementations may use relative consistency proofs, forcing, model construction, proof-theoretic methods, or other appropriate metamathematical techniques.

4 One-Step Agent Action

An agent activation is deliberately abstract.

For an agent g , define

$\text{Step}_g(L_t)$

to be a computation which reads any finite subset of the current bank and returns either no result or a single candidate contribution.

Thus

$$\text{Step}_g(L_t) \in \{\perp\} \cup F,$$

where F is the set of well-formed statements or certificates and $\perp$ denotes failure to produce a contribution on that activation.

If the returned contribution passes the system's verifier, it is deposited into the bank.

Therefore an agent follows the abstract cycle

read bank $\rightarrow$ take one step $\rightarrow$ verify $\rightarrow$ deposit .

The agent may use arbitrarily many previously established premises in taking that step, while depositing at most one new verified result per activation.

5 Global State

Let S_t denote the complete state of the conjecture-settling system at round t .

Abstractly,

$$S_t = (L_t, sP(t), sR(t), sI(t)),$$

where sP , sR , sI represent any private search state maintained by the respective agents.

A transition operator

$$T : S_t \rightarrow S_{t+1}$$

represents one scheduled activation.

For a round-robin scheduler, for example,

$$P, R, I, P, R, I, \dots$$

are activated successively.

The mathematical framework does not require round-robin scheduling. Any fair scheduling rule may be substituted.

6 Settlement Certificates

Define three certificate classes:

Cert+ (C), Cert- (C), Cert0 (C).

They represent respectively:

Cert+ (C) : $A \vdash C$,

Cert- (C) : $A \vdash \neg C$,

and

Cert0 (C) : C is independent of A.

A conjecture is settled when the bank contains a valid member of one of these certificate classes.

7 The Conjecture-Settling Procedure

We may therefore define a partial conjecture-settling function

CSA (C).

Its possible outputs are

$CSA (C) \in \{PROVED, REFUTED, INDEPENDENT\}$.

More precisely,

$CSA (C) = PROVED$

when a valid certificate for $A \vdash C$ is deposited;

$CSA (C) = REFUTED$

when a valid certificate for $A \vdash \neg C$ is deposited; and

$CSA (C) = INDEPENDENT$

when a valid independence certificate is deposited.

The function is partial because an unrestricted mathematical search need not terminate.

8 Stopping Rule

Let

Solved(Lt , C)

mean that Lt contains a valid settlement certificate for C.

The system terminates at the first time

$$\tau = \min\{t : \text{Solved}(L_t, C)\}.$$

At time τ , all agents halt.

Thus the agents compete with respect to settlement objectives while cooperating with respect to mathematical knowledge.

9 Monotonicity

Proposition 1. If statements are added to the bank but never removed, then

$$L_t \subseteq L_{t+1}$$

for every t .

Proof. By definition,

$$L_{t+1} = L_t \cup D_t.$$

Therefore every element of L_t is also an element of L_{t+1} .

10 Soundness of the Bank

Let V be a trusted verification procedure.

A candidate statement x may enter the bank only when

$$V(x) = 1.$$

Consequently, search and verification are separated.

An agent may be heuristic, stochastic, learned, incomplete, or even frequently wrong in its proposals without compromising the logical integrity of the shared bank, provided that the verifier itself is sound.

Proposition 2. Suppose every initial element of L_0 is admissible and every subsequent deposit is accepted only after sound verification. Then every element of every L_t is admissible.

11 Generalization to Many Agents

Nothing essential depends on there being exactly three agents.

Let

$$G = \{g_1, g_2, \dots, g_n\}.$$

Each agent has some objective O_i and transition procedure

Step g_i .

All agents communicate through the same evolving mathematical memory:

$$L_0 \subseteq L_1 \subseteq L_2 \subseteq \dots$$

Different agents may employ entirely different reasoning mechanisms.

For example, one agent could be implemented by a saturation-based automated theorem prover, another by model search, another by an SMT solver, and another by a learned proof-search policy.

The shared-bank abstraction does not depend upon their internal implementations.

12 Example: Law of Excluded Middle
As an initial test conjecture, consider

$$C := \phi \vee \neg\phi.$$

The prover seeks

$$A \vdash \phi \vee \neg\phi.$$

The refuter seeks

$$A \vdash \neg(\phi \vee \neg\phi).$$

The independence agent seeks an appropriate certificate that neither statement follows from the chosen axiom system.

The expected outcome depends critically upon the underlying logic.

For example, the law of excluded middle is valid in classical propositional logic but is not generally derivable in intuitionistic logic.

This illustrates why every conjecture-settling problem must specify not merely a conjecture C , but the formal system A relative to which settlement is sought.

13 Discussion

The essential feature of the framework is not parallelism alone.

Independent proof searches can be dovetailed without communicating. Here, however, successful intermediate mathematical work becomes shared infrastructure.

If agent g_i proves lemma λ , then

$$\lambda \in Lt+1,$$

and every later agent may immediately use λ .

The system therefore combines competition among objectives with cooperation among derivations.

A useful conceptual summary is

many searches + one verified memory + certificate-based stopping .

14 Conclusion

We have defined an abstract multi-agent conjecture-settling process consisting of:

1. a formal background theory,
2. a conjecture,
3. a monotonically growing shared lemma bank,
4. multiple agents pursuing distinct settlement objectives,
5. a trusted verification mechanism,

6. a scheduler, and

7. a certificate-based stopping condition.

The framework deliberately leaves the internal reasoning algorithm of each agent unspecified.

This permits automated theorem provers, model finders, proof assistants, symbolic algorithms, and learned reasoning systems to be substituted without changing the abstract conjecture-settling architecture.